

How to Generate Supabase API & Firebase API

Complete Technical Integration & Key Management Guide

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1. How to Generate Supabase API Credentials

Supabase is a modern open-source Firebase alternative built on PostgreSQL. Follow these exact steps to obtain your project URL, anon key, and service_role key:

Step 1.1: Create a Supabase Account & Project

- 1. Navigate to **https://supabase.com** and click **Sign in** or **Start your project**.
- 2. Authenticate using your GitHub or Google account.
- 3. In the Supabase Dashboard, click **New Project**.
- 4. Fill in the project configuration:
 - **Name:** e.g., *agnexa-production-db*
 - **Database Password:** Generate and save a strong password.
 - **Region:** Select the closest data center (e.g., *ap-south-1 (Mumbai)* for India).
- 5. Click **Create new project** and allow 1–2 minutes for database provisioning.

Step 1.2: Locate & Copy Your API Keys

- 1. On your project dashboard, look at the bottom-left sidebar and click the **Project Settings (Gear Icon)**.
- 2. In the Settings sub-menu, select **API** (under the Configuration section).
- 3. You will see two primary sections:

Parameter	Description	Where to Use
Project URL	Base REST and Realtime gateway URL (e.g., <code>https://xyz.supabase.co</code>)	Backend & Frontend API client initialization
anon / public	Anonymous public client key. Adheres strictly to Row Level Security (RLS) policies.	Frontend React apps or public API calls
service_role / secret	Superuser secret key. Bypasses all Row Level Security policies.	Backend only (.env). Never expose to clients!

Step 1.3: Apply the Database Schema & Tables

- 1. Click **SQL Editor** in the left navigation sidebar.
- 2. Click **New query**.
- 3. Paste the provided SQL file from `server/src/database/supabase_schema.sql`.
- 4. Click **Run (Cmd+Enter / Ctrl+Enter)**. This automatically generates all tables (services, users, contacts, portfolio, blogs) with seed data!

```
# Agnexa server/.env configuration
SUPABASE_URL=https://your-project-ref.supabase.co
SUPABASE_ANON_KEY=eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
SUPABASE_SERVICE_ROLE_KEY=eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
```

2. How to Generate Firebase API Credentials

Google Firebase provides two types of API credentials depending on whether you are integrating a web client or a secure backend:

Step 2.1: Create a Project in Firebase Console

- 1. Go to <https://console.firebase.google.com/> and sign in with your Google Account.
- 2. Click **Add project** (or *Create a project*).
- 3. Enter your project name (e.g. *Agnexa-Platform*), accept terms, and click **Continue**.
- 4. Choose whether to enable Google Analytics, then click **Create project**.

Step 2.2: Generate Web Client API Credentials (For Frontend)

- 1. In your Project Overview screen, click the **Web icon** (</>) to register an app.
- 2. Enter an App nickname (e.g., *agnexa-web-portal*) and click **Register app**.
- 3. Firebase will immediately display the configuration object containing your Web API Key:

```
// Firebase Web Client Config (client/.env or firebaseConfig.js)
const firebaseConfig = {
  apiKey: 'AIzaSyA8_exampleApiKey94827',
  authDomain: 'agnexa-platform.firebaseio.com',
  projectId: 'agnexa-platform',
  storageBucket: 'agnexa-platform.appspot.com',
  messagingSenderId: '829471920481',
  appId: '1:829471920481:web:d8471932948b29'
};
```

Step 2.3: Generate Admin SDK Service Account JSON (For Backend)

- 1. Click the **Gear icon (Project Settings)** at the top left of the Firebase Console.
- 2. Navigate to the **Service accounts** tab.
- 3. Select **Node.js** and click the blue button: **Generate new private key**.
- 4. Confirm by clicking **Generate key**. A .json file will download to your computer.
- 5. Save this file as `serviceAccountKey.json` inside your backend directory (add it to `.gitignore`!).

```
// Backend Firebase Admin SDK Initialization (server/src/config/firebase.js)
import admin from 'firebase-admin';
import serviceAccount from './serviceAccountKey.json' assert { type: 'json' };

admin.initializeApp({
  credential: admin.credential.cert(serviceAccount)
});
export const firestore = admin.firestore();
```

3. Supabase vs Firebase Architecture Comparison

Feature	Supabase	Firebase
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Database Engine	PostgreSQL (Relational, SQL, pgvector for AI)	Cloud Firestore / Realtime DB (NoSQL Documents)
Query Flexibility	Full SQL joins, aggregates, stored procedures, triggers	Shallow document reads, composite index constraints
Security Enforcement	Row Level Security (RLS) defined directly in SQL	Firebase Security Rules written in JSON/DSL syntax
Self-Hosting & Portability	100% open source, run locally via Docker with 0 lock-in	Proprietary Google Cloud ecosystem

Crucial Security Principle:
Never commit your Supabase service_role key or Firebase serviceAccountKey.json into public Git repositories. Always store them in .env files or cloud secret managers.